

# Experiment 3

## 1. Aim

Implement control structures and form validation; create a grading system based on user-entered marks.

## 2. Software/Tools Required

- Visual Studio Code (VS Code)
- HTML
- JavaScript

## 3. Code

```
<!DOCTYPE html>
<html>
  <head>
    <title> Student Grading System </title>
  </head>
  <body>
    <h2>
      Shravasti's Student Grading System (24070521194):
    </h2>
    Enter Marks:
    <input type = "number" id = "marks">
    <button onclick = "checkGrade()">Submit</button>
    <p id = "result"></p>
    <script>
      function checkGrade()
      {
        var marks = document.getElementById("marks").value;

        if (marks == "")
        {
          alert("Please Enter Marks.");
          return;
        }
        if (marks < 0 || marks > 100)
        {
          alert ("Enter marks between 0 and 100");
          return;
        }
      }
    </script>
  </body>
</html>
```

```
if (marks >= 90)
    grade = "A";
else if (marks >= 75)
    grade = "B";
else if (marks >= 60)
    grade = "C";
else if (marks >= 40)
    grade = "D";
else
    grade = "Fail";

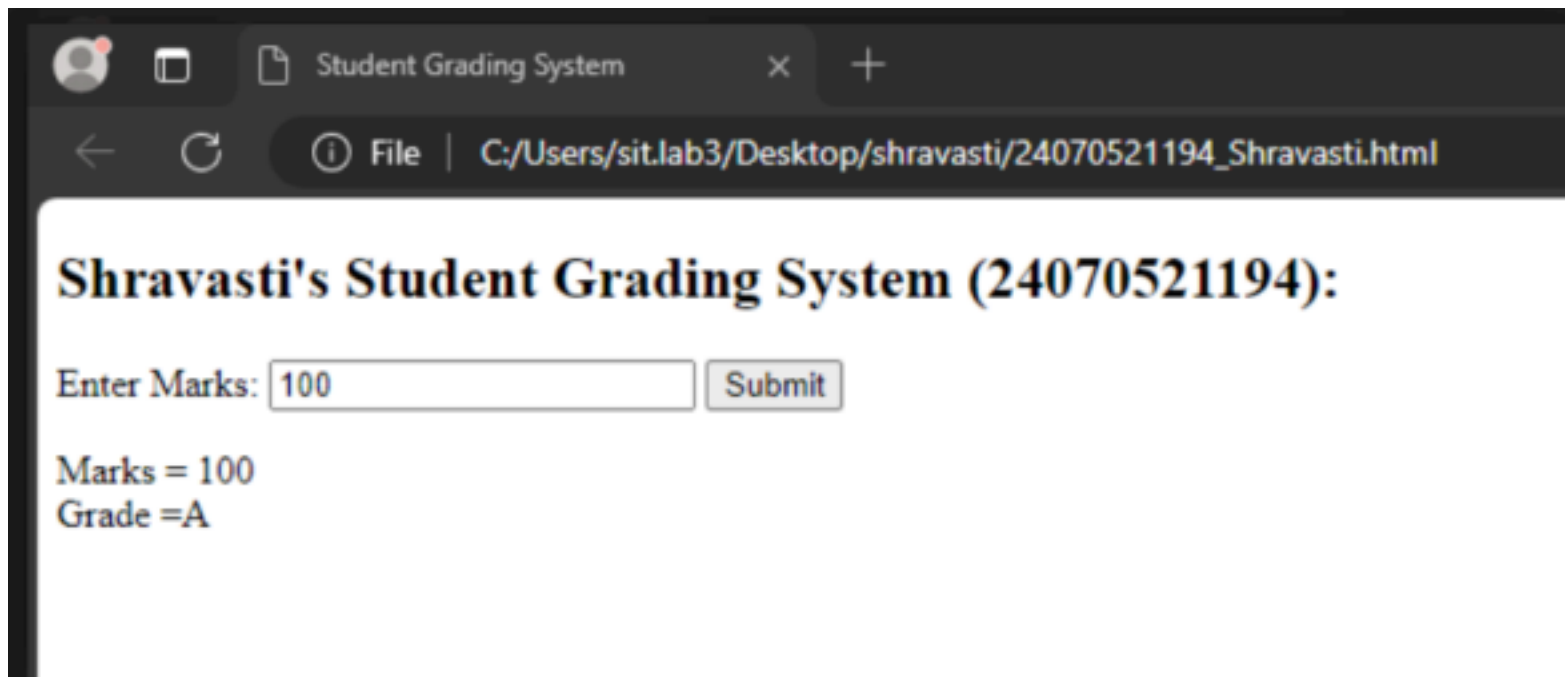
document.getElementById ("result").innerHTML = "Marks = " + marks + "<br>Grade =" + grade;
}
</script>
</body>
</html>
```

## 4.Theory

Control structures in JavaScript, such as **if**, **if-else**, and **else-if**, are used to make decisions based on conditions. Form validation ensures that users enter valid data before processing it.

In this experiment, the user enters marks through a form. JavaScript validates the input to ensure the marks are between **0 and 100**. Based on the entered marks, the program uses control structures to assign an appropriate grade and display the result.

## 5. Output



## 6. Result/Conclusion

The experiment was executed successfully. The concepts of JavaScript control structures and form validation were implemented to accept user-entered marks, validate the input, and display the appropriate grade based on the entered marks.